VOLTAGE-DIVIDER BIAS

E-MOSFET DC BIASING

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

EXAM

50%

LABORATORY

PENALTY

0

ABSENT

3

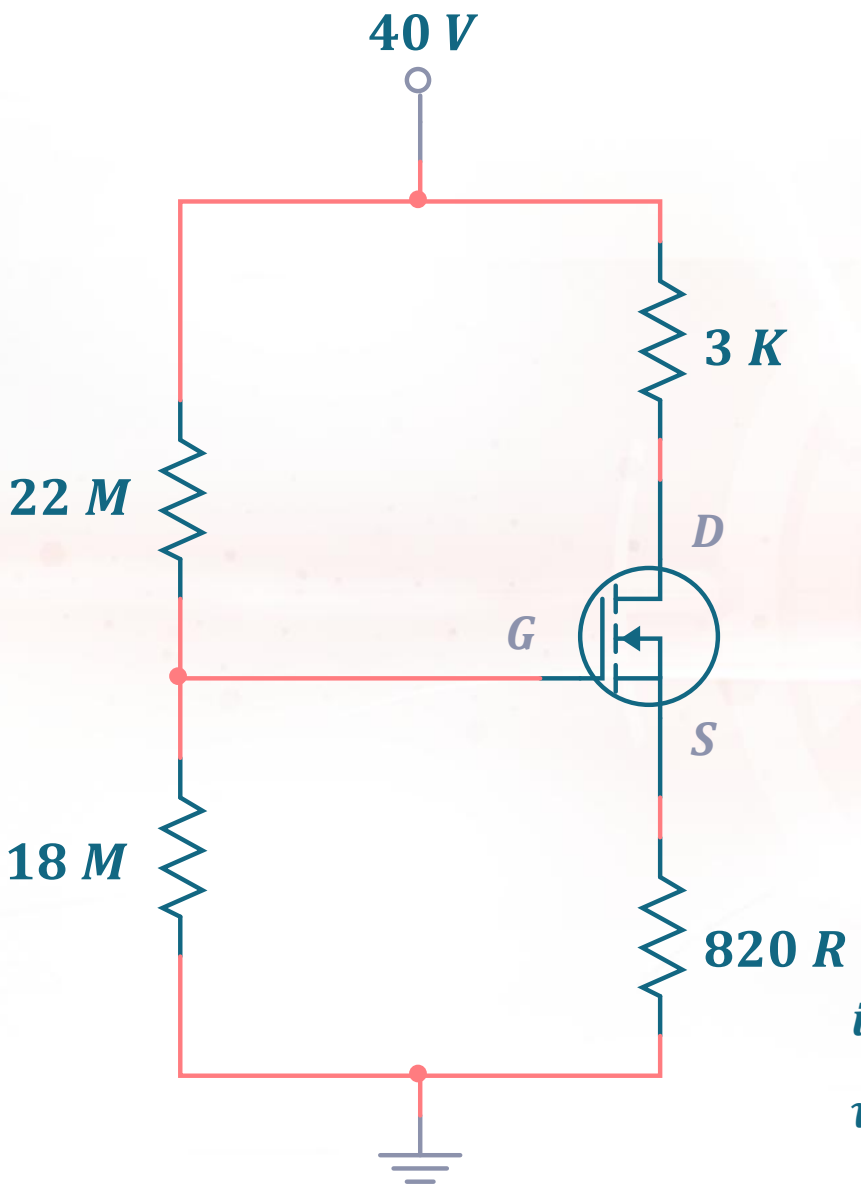
45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Voltage-Divider Bias Configuration

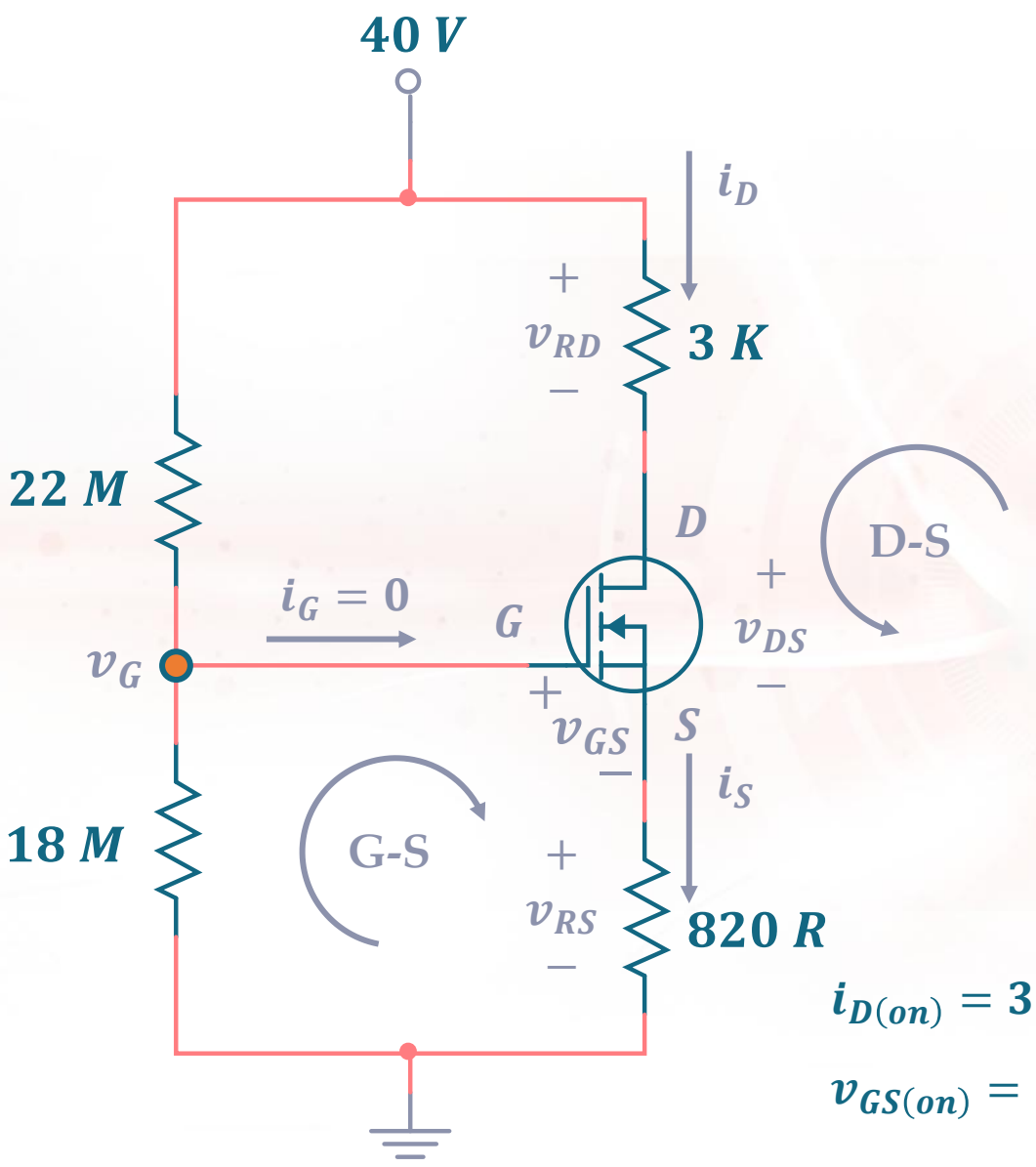


$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$v_{GS(TH)} = 5\ V$

For the given **voltage-divider** biasing arrangement, determine i_{DQ}, v_{GSQ} and v_{DSQ} for the enhancement-type MOSFET. **Sketch** the transfer characteristics for $v_{GS} = 8\ V$ and $v_{GS} = 15\ V$.



$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$v_{GS(TH)} = 5\ V$

Gate Voltage

$$v_G = v_{DD}\frac{R_2}{R_1 + R_2}$$

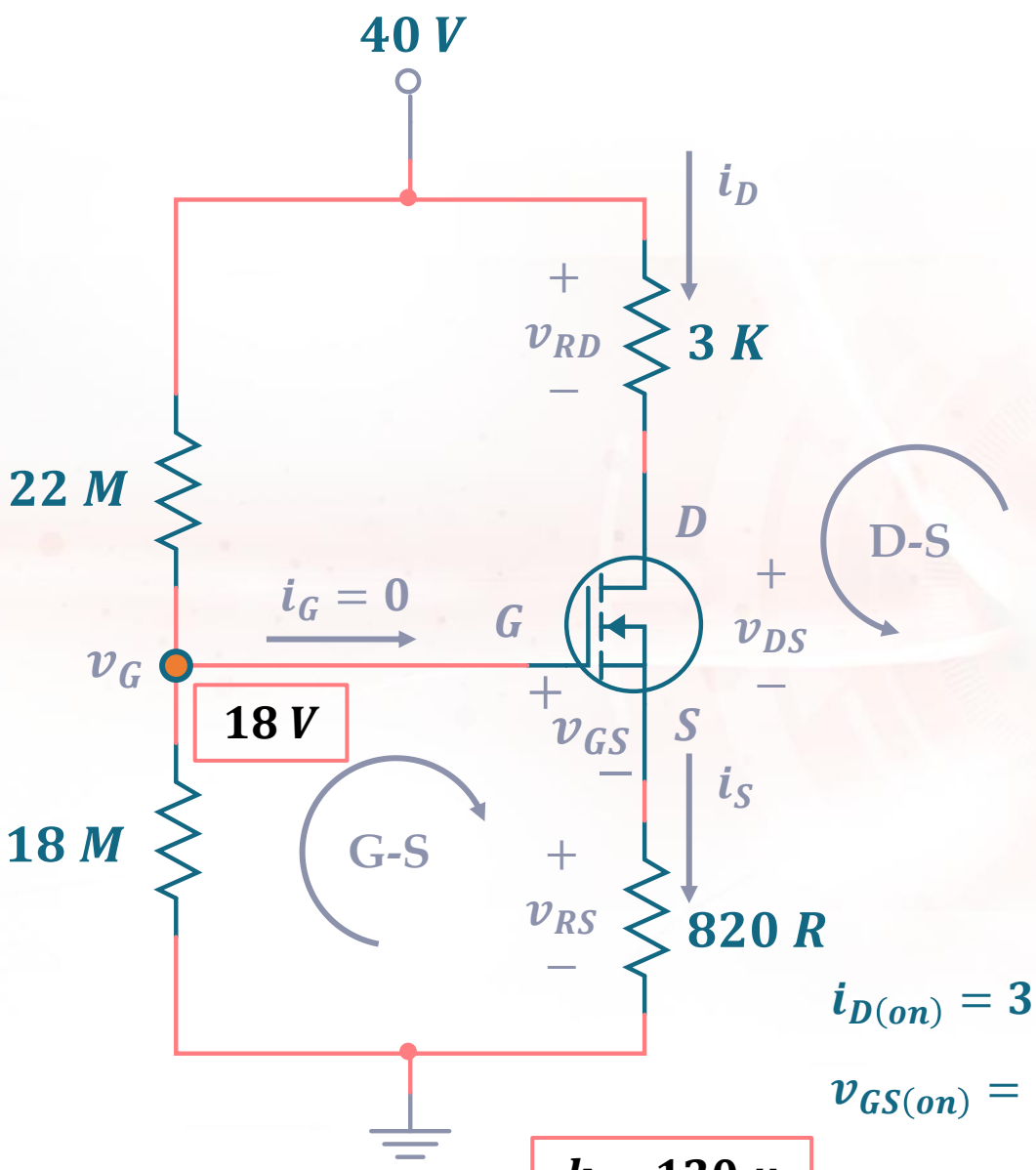
$$v_G = 40\left(\frac{18\ M}{22\ M + 18\ M}\right)$$

$$v_G = 18\ V$$

$$k = \frac{i_{D(on)}}{\left(v_{GS(on)} - v_{GS(th)}\right)^2}$$

$$k = \frac{3\ mA}{(10 - 5)^2}$$

$$k = 120\ \mu A/V^2$$



$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$k = 120\ \mu$ $\quad v_{GS(TH)} = 5\ V$

KVL @ G-S

$$-v_G + v_{GS} + v_{RS} = 0$$

$$v_{GS} = v_G - v_{RS}$$

$$v_{GS} = v_G - i_D R_S$$

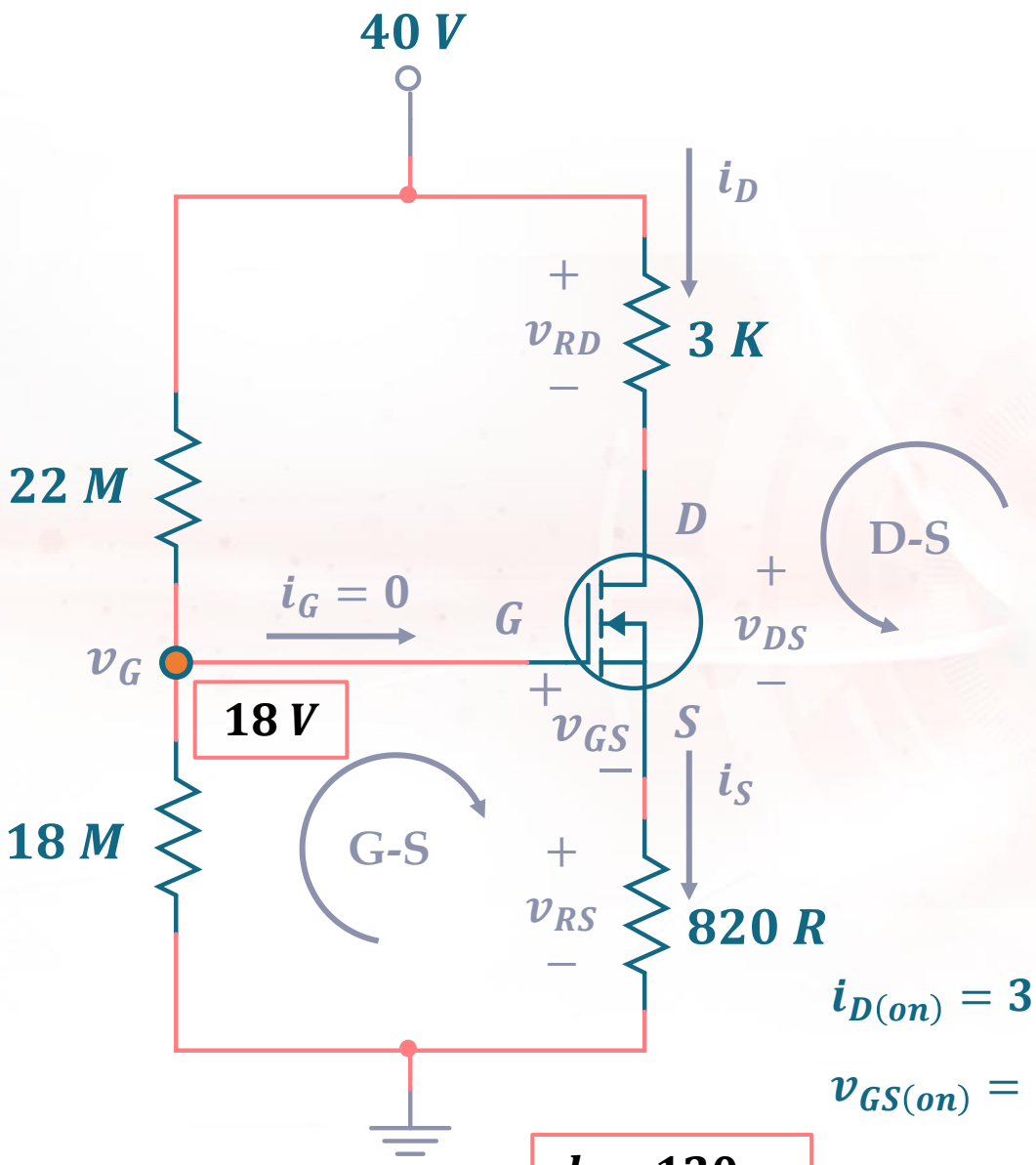
$$v_{GS} = v_G - k\left(v_{GS} - v_{GS(th)}\right)^2 R_S$$

$$v_{GS} = 18 - 120\ \mu\ (v_{GS} - 5)^2 (820)$$

$$v_{GS} = 18 - 98.4\ m\ \left(v_{GS}^2 - 10 v_{GS} + 25\right)$$

$$v_{GS} = 18 - 98.4\ m v_{GS}^2 + 984\ m v_{GS} - 2.46$$

$$0 = -98.4\ m v_{GS}^2 - 16\ m v_{GS} + 15.54$$



$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$k = 120\ \mu$ $\quad v_{GS(TH)} = 5\ V$

KVL @ G-S

$$0 = -98.4\ mv_{GS}^2 - 16\ mv_{GS} + 15.54$$

$$\boxed{v_{GSQ} = 12.49\ V}$$

ans

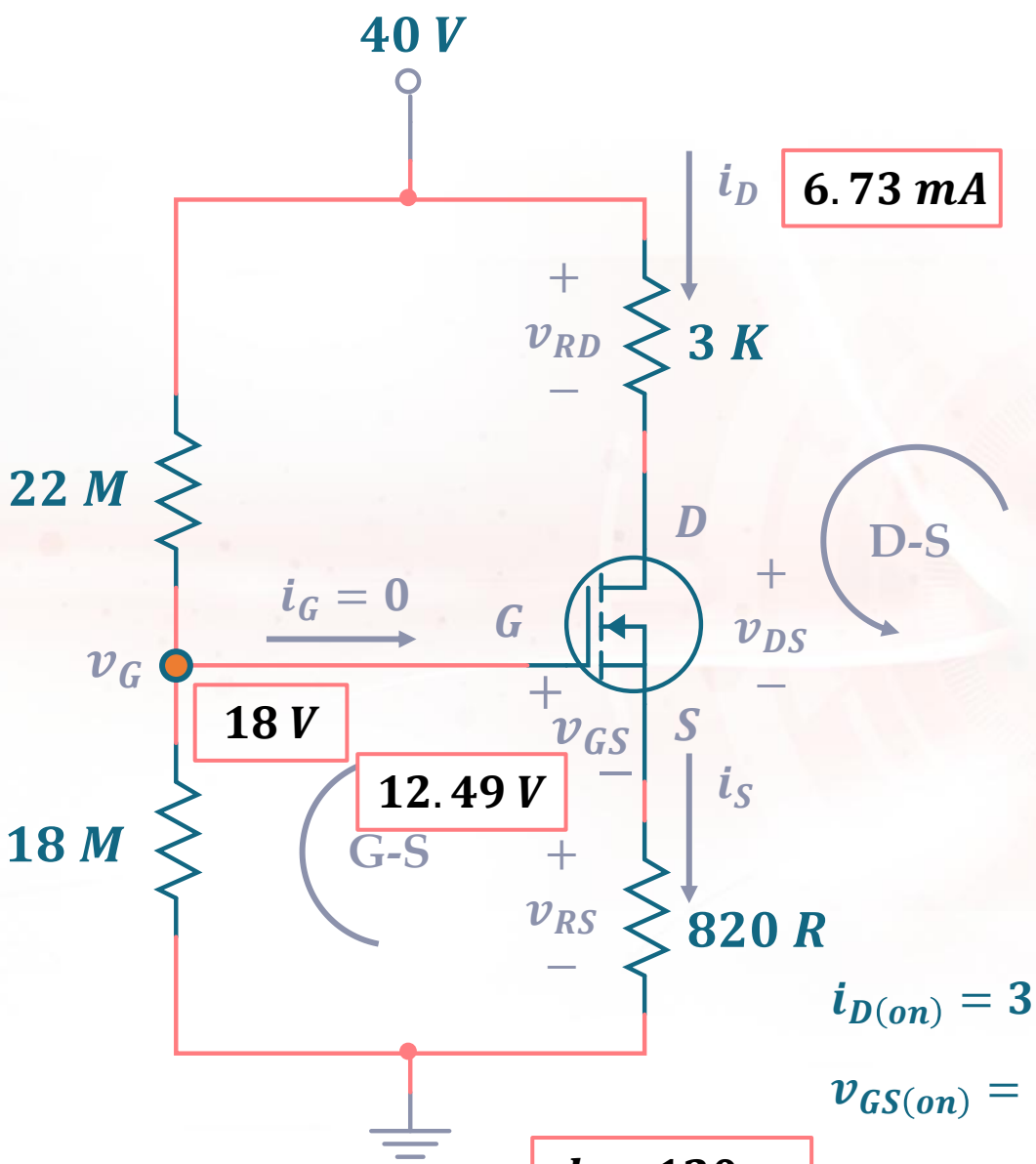
$$v_{GS} = -12.65\ V$$

$$i_D = k\left(v_{GS} - v_{GS(th)}\right)^2$$

$$i_D = 120\ \mu(12.49 - 5)^2$$

$$\boxed{i_{DQ} = 6.73\ mA}$$

ans



$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$k = 120\ \mu$ $\quad v_{GS(TH)} = 5\ V$

KVL @ D-S

$$-v_{DD} + v_{RD} + v_{DS} + v_{RS} = 0$$

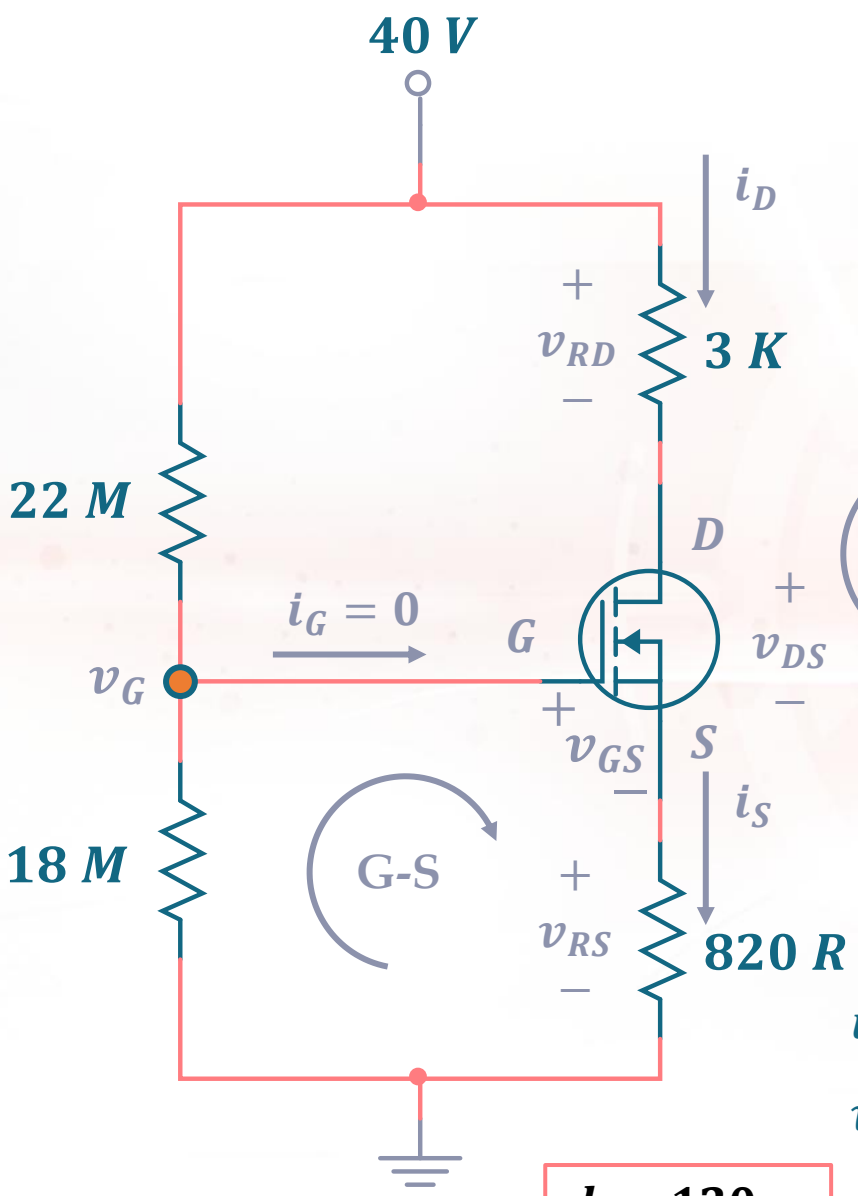
$$v_{DS} = v_{DD} - v_{RD} - v_{RS}$$

$$v_{DS} = v_{DD} - i_D(R_D + R_S)$$

$$v_{DS} = 40 - 6.73\ mA(3K + 820)$$

$$v_{DSQ} = 14.29\ V$$

ans



$i_{D(on)} = 3\ mA$

$v_{GS(on)} = 10\ V$

$k = 120\ \mu$ $\quad v_{GS(TH)} = 5\ V$

Transfer Characteristics

$$i_D = k\left(v_{GS} - v_{GS(th)}\right)^2\Big|_{v_{GS}=8\ V}$$

$$i_D = 120\ \mu(8-5)^2\Big|_{v_{GS}=8\ V}$$

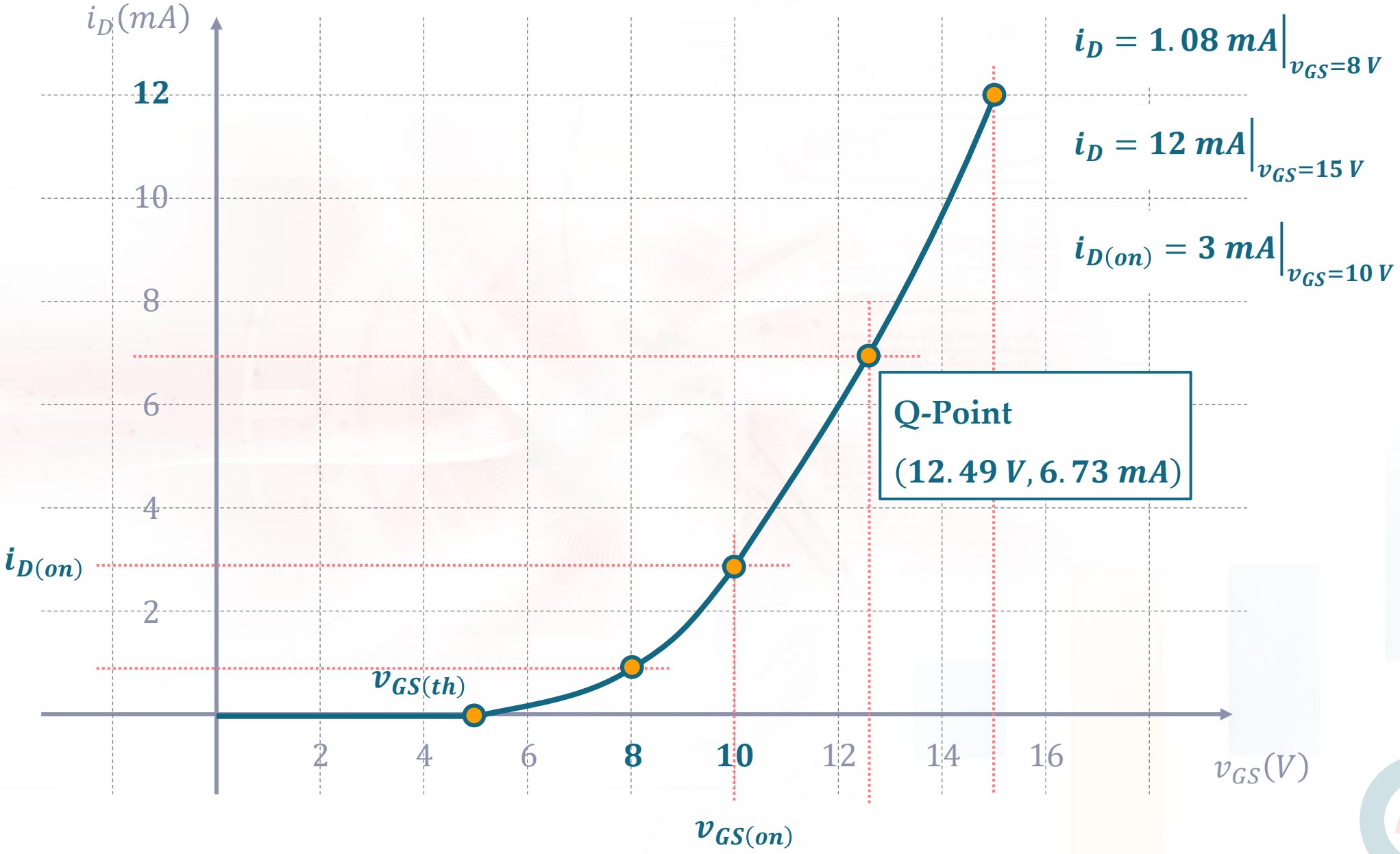
$$i_D = 1.08\ mA\Big|_{v_{GS}=8\ V}$$

$$i_D = k\left(v_{GS} - v_{GS(th)}\right)^2\Big|_{v_{GS}=15\ V}$$

$$i_D = 120\ \mu(15-5)^2\Big|_{v_{GS}=15\ V}$$

$$i_D = 12\ mA\Big|_{v_{GS}=15\ V}$$

TRANSFER CHARACTERISTICS



LABORATORY